

# REECE DICKINSON

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## EDUCATION

### Cal Poly Humboldt — B.S. in Computer Science

Expected May 2028

- **GPA: 3.94** | Presidential Scholar | Dean's List (Every Term)
- **Relevant Coursework:** Systems Programming (CS 375), Data Structures (CS 211), OOP (CS 112), Web-Apps Using Databases (CS 328), Database Design (CS 325), Computer Architecture (CS 243)

## TECHNICAL SKILLS

- **Languages & Tools:** Python (FastAPI), C/C++, Java, SQL, React, JavaScript/TypeScript, Node.js, ComfyUI, Git/GitHub
- **Systems & AI/ML:** Full-Stack GenAI, Retrieval-Augmented Generation (RAG), Local Diffusion Pipeline Engineering (Wan 2.2, LTX-2.3, Chroma), ComfyUI Pipeline Optimization, POSIX System Calls (fork, exec, wait), PGVector, Supabase, Alembic
- **Certifications & Languages:** Anthropic AI Fluency, Anthropic Claude 101 | Bilingual (Fluent English, Conversational Spanish)

## TECHNICAL PROJECTS

### Full-Stack RAG Q&A — Python, React, Supabase, PGVector

Jun 2026 – Present

- Architected an end-to-end investment research chat application with a real-time streaming layer.
- Designed a robust PostgreSQL data model with PGVector embeddings and Alembic migrations.
- **Trust Contract:** Implemented a page-level citation pipeline to guarantee grounded, verifiable responses.
- Full system architecture, schema designs, and live Railway deployment demo available on portfolio.

### Custom UNIX Shell — C, POSIX, Linux Systems

Fall 2026 – Present

- Developing a custom, POSIX-style shell interpreter in C to handle system-level process execution.
- Managing parent-child process lifecycles via POSIX system calls (fork, exec, wait) and custom signal handlers (SIGINT).
- Process diagrams, memory leak profiling (Valgrind), and source code available on portfolio.

### Local GenAI Pipeline — ComfyUI, PyTorch, RTX 4090

Summer 2026 – Present

- Deployed open-weight diffusion models (Chroma1-HD, Wan 2.2 14B MoE, LTX-2.3 22B) on an RTX 4090 workstation, configuring FP8 quantization, VAE selection, and T5-XXL/Gemma 3 text-encoder pairings.
- Diagnosed and patched cascading Python dependency conflicts in third-party ComfyUI nodes (kornia API breaks, huggingface\_hub deprecations) and resolved a vendored BigVGAN vocoder bug via source code inspection.
- Engineered an audio-synchronized video pipeline (Wan 2.2 → MMAudio) by fixing frame-rate truncation bugs and resolving file-handoff race conditions using pure in-memory node connections.
- Node-graph execution architectures, custom Python patch logs, and workflow demos available on portfolio.

### LeetCode 75 & Homelab — Algorithms & Linux Admin

Summer 2026 – Ongoing

- Solved all 75 Grind 75 problems to master core data structures and algorithmic patterns.
- Administer multi-user Linux game servers, debugging networking ports and mod compatibility.

## PROFESSIONAL EXPERIENCE

### Cal Poly Humboldt Library — Scholar Intern (Terraform)

Feb 2026 – May 2026

- Rebuilt a physical AR sandbox from scratch, routing Kinect depth camera inputs to an AI StreamDiffusion pipeline.
- Collaborated across IT and management teams to resolve architecture bugs, releasing the code under an MIT License.
- Amsterdam-guided design brief, hardware wiring guides, and full reflection available on portfolio.

### Ramy Dickinson Construction — Contractor Assistant

Summer 2023 – Present

- Manage materials, job-site logistics, and daily operations requiring independent, real-time decision-making.

## LEADERSHIP & COMMUNITY

- **President, Spanish National Honor Society:** Organize community initiatives to bridge English- and Spanish-speaking members.
- **Aviation Club Webmaster:** Built and optimized the club website to support recruitment and community airport events.
- **Community Volunteer:** Over 100 volunteer hours with the Veterans' Hall, Trails Steward, and Trinidad Coastal Land Trust.